

(Compare this to Exercise 0.9!) Note that in fixed dimension, $f_{d-1}(C_d(n))$ grows like a polynomial of degree $\lfloor \frac{d}{2} \rfloor$ in the number of vertices.

Here is a brief asymptotic argument, due to Seidel [490] (see also Mulmuley [417]), for this corollary to the upper bound theorem. Namely, consider any shelling of ∂P . For every facet we get that either the restriction R_j or its complement $F_j \setminus R_j$ has size at most $\lfloor \frac{d}{2} \rfloor$. So, either in the shelling or in its reverse we have that F_j has a restriction of size at most $\lfloor \frac{d}{2} \rfloor$, and the restriction sets in a shelling are distinct by construction. Thus the number of facets is at most twice the number of k -faces of P with $k \leq \lfloor \frac{d}{2} \rfloor$. From this we get

$$f_{d-1} \leq 2 \sum_{i=0}^{\lfloor \frac{d}{2} \rfloor} \binom{n}{i},$$

and this rough estimate bounds f_{d-1} by a polynomial of degree $\lfloor \frac{d}{2} \rfloor$ in n .

8.5 Some Extremal Set Theory

We have used already that the simplicial complex $\mathcal{C}$ with n vertices can be identified with a *set system*, the collection of subsets $\mathcal{S}(\mathcal{C})$ of an n -set,

$$\mathcal{S}(\mathcal{C}) := \{\text{vert}(G) : G \in L(\mathcal{C})\}.$$

For the following we identify the vertex set of $\mathcal{C}$ with the set

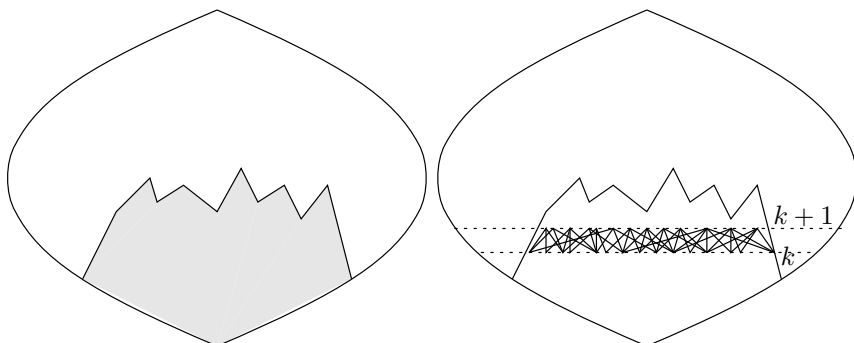
$$[n] := \{1, 2, \dots, n\},$$

and the k -faces of the complex with the $(k+1)$ -subsets of the ground set $[n]$, for $-1 \leq k \leq \dim(\mathcal{C})$. Thus, if $\mathcal{C}$ is a pure $(d-1)$ -dimensional simplicial complex, then it is determined by its family of facets, which is a subset of $\binom{[n]}{d}$, the collection of d -subsets of $[n]$.

The construction behind this identifies *geometric simplicial complexes*, as we get them for example as boundary complexes of simplicial polytopes, with *abstract simplicial complexes*, where we only retain the information on the vertex set, and the information about “which vertex sets correspond to faces, respectively facets.” This approach is useful for all problems that are not concerned with the geometry of a complex, but only with its combinatorial structure. The combinatorial structure of a complex, however, is *faithfully* represented by the abstract set system: from the set system data, it is easy to reconstruct the simplicial complex (this is a process known as *geometric realization* of the abstract simplicial complex).

On the next page is a “generic picture” of a simplicial complex, viewed as a set system. (Of course, this need not be *your* way of viewing set systems — supply your own sketch!) The left version just shows you the “shape” of a simplicial complex within the lattice of all subsets of $[n]$, while the

right side shows you the bipartite graph of all k -faces and $(k+1)$ -faces, that is, of all subsets of size $k+1$ and k within the complex. In both cases the minimal element is $\emptyset$, which is always supposed to be contained in the complex, corresponding to $f_{-1} = 1$.



Lemma 8.29. (Sperner [509]) *If $\mathcal{C}$ is a simplicial complex of dimension d on $f_0 \leq n$ vertices, then for $0 \leq k \leq d$ one has*

$$\frac{f_k}{f_{k-1}} \leq \frac{n-k}{k+1} = \frac{\binom{n}{k+1}}{\binom{n}{k}},$$

with equality if and only if $\mathcal{C}$ is k -neighborly, such that $f_k = \binom{n}{k+1}$ and $f_{k-1} = \binom{n}{k}$.

Proof. We double-count the edges in the bipartite graph above. Every $(k+1)$ -set contains $k+1$ different k -sets, thus there are $(k+1)f_k$ edges. Similarly, every k -set in the complex is contained in at most $n-k$ different $(k+1)$ -sets of the complex, so there are not more than $(n-k)f_{k-1}$ edges, and we get the inequality $(k+1)f_k \leq (n-k)f_{k-1}$.

If we have equality, then with every k -set the complex contains all the $(k+1)$ -sets that contain it. However, since we can get from every $(k+1)$ -set to every other one by throwing out one element, adding a new one, throwing one out, and so on, this means that in the equality case the complex contains all the $(k+1)$ -sets, or none. $\square$

The boundary case $k = 0$ is trivial. For $k = 1$, Sperner's lemma just says that on n vertices there cannot be more than $\binom{n}{2}$ edges. For higher k it gets more interesting. Here is one thing we can easily derive from it.

Lemma 8.30. *Let $\mathcal{C}$ be a simplicial complex of dimension $d-1$, with $n = f_0(\mathcal{C})$ vertices. If $n \geq dk - (k-1)^2$, then the h -vector of $\mathcal{C}$ satisfies*

$$h_k \leq \binom{n-d-1+k}{k},$$

with equality if and only if $\mathcal{C}$ is k -neighborly, that is, if $f_{k-1} = \binom{n}{k}$.

Proof. We group the terms of $h_k = \sum_{i=0}^k (-1)^i \binom{d-k+i}{i} f_{k-1-i}$ in pairs, as

$$\begin{aligned} h_k &= \binom{d-k+0}{0} f_{k-1} - \binom{d-k+1}{1} f_{k-2} \\ &+ \binom{d-k+2}{2} f_{k-3} - \binom{d-k+3}{3} f_{k-4} \\ &+ \dots \end{aligned}$$

Each of the terms is of the form $\binom{d-j-1}{k-j-1} f_j - \binom{d-j}{k-j} f_{j-1}$ for $1 \leq j \leq k-1$. (If k is odd we'll have an extra term of $\binom{d}{k} f_{-1}$ at the end, but that's a constant anyway.) By Sperner's lemma, we can bound this by

$$\begin{aligned} \binom{d-j-1}{k-j-1} f_j - \binom{d-j}{k-j} f_{j-1} &= \binom{d-j-1}{k-j-1} f_{j-1} \left\{ \frac{f_j}{f_{j-1}} - \frac{d-j}{k-j} \right\} \\ &\leq \binom{d-j-1}{k-j-1} f_{j-1} \left\{ \frac{n-j}{j+1} - \frac{d-j}{k-j} \right\}. \end{aligned}$$

This is maximized exactly if $f_{j-1} = \binom{n}{j}$ for all j , provided that we are sure that the differences $\frac{n-j}{j+1} - \frac{d-j}{k-j}$ are nonnegative. From this we get that h_k is maximized by the simplicial complexes that are k -neighborly, if we have

$$n \geq j + (j+1) \frac{d-j}{k-j} \quad \text{for } j = 1, \dots, k-1.$$

Since this lower bound is monotone in j , we only need to consider this for $j = k-1$, and get the condition $n \geq (k-1) + k(d-k+1)$, which we had required to hold. $\square$

Note that this lemma is *false* without the assumption that n is large enough: for the complex in Example 8.20(iii) we have $d = 6$, $f_0 = n = 12$, and

$$h_3 = 60 > 56 = \binom{8}{3} = \binom{n-d-1+3}{3}.$$

From this elementary lemma we get McMullen's Lemma 8.26, and thus a proof of the upper bound theorem, for polytopes with a sufficiently large number of vertices. This simple proof not only works for polytopes: the argument equally applies to all kinds of simplicial complexes that satisfy the Dehn-Sommerville equations. This includes all *spherical polytopes* (corresponding to simplicial fans, see Kleinschmidt & Smilansky [337]), and even more generally, all simplicial Eulerian pseudomanifolds (see Klee [323], Bayer & Billera [62], Chan, Jungreis & Stong [146], and Stanley [517, Sect. 3.14]).

Corollary 8.31 (The upper bound theorem for complexes with many vertices). *Let C be a $(d-1)$ -dimensional simplicial complex that satisfies the Dehn-Sommerville equations $h_k = h_{d-k}$.*

If the number $n = f_0$ of vertices satisfies

$$n \geq d \lfloor \frac{d}{2} \rfloor - (\lfloor \frac{d}{2} \rfloor - 1)^2,$$

then for $0 \leq k < d$ the complex cannot have more k -faces than the boundary of the cyclic polytope $C_d(n)$:

$$f_k(\mathcal{C}) \leq f_k(C_d(n)).$$

Proof. We get $h_k \leq \binom{n-d-1+k}{k} = h_k(C_d(n))$ for $k \leq \lfloor \frac{d}{2} \rfloor$ from Lemma 8.30, and for $k > \lfloor \frac{d}{2} \rfloor$ from the Dehn-Sommerville equations. The rest follows from the fact that the f_k s are positive combinations of the h_i s with $i \leq k+1$. $\square$

In this proof, you can see some of the power of the translation of geometric simplicial complexes into finite set systems (abstract simplicial complexes). In this setting, extremal problems about simplicial complexes are a principal topic of “extremal set theory.” We will review in the rest of this section some basic concepts, constructions, and results from this field — some of them without proofs, to save time and space. You might want to look at the wonderful survey by Greene & Kleitman [241] for some of the missing details.

A basic tool of extremal set theory is the use of various partial and linear orderings on the k -subsets of an n -set (i.e., on the $(k-1)$ -faces of a complex). Since we assume that the vertex set is $[n]$, that is, the vertices are labeled $1, 2, \dots$, there is a natural linear ordering (“well-ordering”) on the vertex set. With this, we can in particular talk about the largest element $\max(G)$, if G is nonempty.

Using this, we define the *r-lex order* (or *reverse lexicographic ordering*) on the k -subsets of vertices. For this we write

$$G \prec H$$

if and only if $G \neq H$ and the largest element in which G and H differ is in H , that is, if

$$\max(G \setminus H) < \max(H \setminus G).$$

Equivalently, this means that either $\max(G) < \max(H)$, or $\max(H) = \max(G) =: p$ and $G \setminus p \prec H \setminus p$.

In the definition of the r-lex order, the number of elements n is not specified. Thus, we can take “ $\prec$ ” as a linear order on the set of all the k -subsets of $\mathbb{N}$. Furthermore, for every k -subset $G \subset \mathbb{N}$, there is only a finite number of k -subsets of $\mathbb{N}$ that are smaller than G , because $G \succ H$ implies that $H \in \binom{[n]}{k}$ for $n := \max(G)$. This means that we can use the r-lex order to enumerate all the k -subsets of $\mathbb{N}$, as $F_1(k), F_2(k), \dots$. So, we define $F_j(k)$ to be the j th subset in this increasing listing according to r-lex order.

For example, the r-lex order on the 3-subsets of $\mathbb{N}$ begins

$$123 \prec 124 \prec 134 \prec 234 \prec 125 \prec 135 \prec 235 \prec 145 \prec \dots,$$

and with the above notation, this list shows

$$F_1(3) \prec F_2(3) \prec F_3(3) \prec F_4(3) \prec F_5(3) \prec F_6(3) \prec F_7(3) \prec F_8(3) \prec \dots$$

For $n, k \geq 0$, there is a unique *binomial expansion* of n of the form

$$n = \binom{a_k}{k} + \binom{a_{k-1}}{k-1} + \dots + \binom{a_2}{2} + \binom{a_1}{1}$$

with $a_k > a_{k-1} > \dots > a_2 > a_1 \geq 0$.

In fact, existence and uniqueness of this expansion are easy to verify, by choosing a_k first, a_{k-1} after that, and so on. A more systematic explanation may be the following. Define the integers $a_k > a_{k-1} > \dots > a_2 > a_1 \geq 0$ by setting

$$F_{n+1}(k) =: \{a_1 + 1, a_2 + 1, \dots, a_{k-1} + 1, a_k + 1\}_<.$$

(Here the subscript “<” indicates that the elements are listed in increasing order.) Then there are exactly n different k -subsets $G \subset \mathbb{N}$ that are smaller than $\{a_1 + 1, \dots, a_k + 1\}$ in r-lex order. Namely, $\binom{a_k}{k}$ of them have a maximal element smaller than $a_k + 1$; $\binom{a_{k-1}}{k-1}$ have maximal element $a_k + 1$ but the next smallest element smaller than $a_{k-1} + 1$; and so on.

One more thing is easy to see: the $(k-1)$ -subsets of $\mathbb{N}$ that are contained in some $F_j(k)$ with $j \leq n+1$ also have maximal element smaller than $a_k + 1$, or they have maximal element $a_k + 1$ but the next element is smaller than $a_{k-1} + 1$, etc. — so there are exactly

$$\partial_k(n+1) := \binom{a_k}{k-1} + \binom{a_{k-1}}{k-2} + \dots + \binom{a_2}{1} + \binom{a_1}{0}$$

$(k-1)$ -subsets contained in the k -sets $F_1(k), \dots, F_{n+1}(k)$.

For an example, let $k = 3$ and $n = 7$. We expand

$$7 = \binom{4}{3} + \binom{3}{2} + \binom{0}{1},$$

and from this we see that $F_8(3) = \{5, 4, 1\}$, consistent with our listing above. There are 7 smaller sets in r-lex order, where $4 = \binom{4}{3}$ have largest element smaller than 5; $3 = \binom{3}{2}$ have largest element 5 but the next element smaller than 4; and $0 = \binom{0}{1}$ have the two largest elements 5 and 4 but the smallest element smaller than 1 (impossible). Also, there are

$$\partial_3(8) = \binom{4}{2} + \binom{3}{1} + \binom{0}{0}$$

2-subsets contained in the first eight 3-sets, namely $6 = \binom{4}{2}$ with largest element smaller than 5; $3 = \binom{3}{1}$ with largest element 5 but the next one smaller than 4; and $1 = \binom{0}{0}$ with the elements 5 and 4. Note that this last one is contained in $F_8(3)$, but not in a smaller 3-set.

The r -lex order is very natural in various respects. For example, it yields a shelling order for the $(k-1)$ -skeleton of the simplex Δ_d , for $k \leq n = d+1$ (Exercise 8.24(i)). In fact, many other linear orders work as well. However, it is a fascinating open problem whether skeleta of simplices are extendably shellable; see Problem 8.24(iii)*.

Perhaps the most basic result of extremal set theory, and a principal application of r -lex order, is the characterization of the f -vectors of simplicial complexes. It is known as the Kruskal-Katona theorem [344] [318], although Schützenberger [485] was earlier, and even before this Harper got close: his paper [271] does not explicitly state the theorem, but the result is easy to derive, and I was told that Harper was aware of it at the time.

Theorem 8.32 (Kruskal-Katona theorem).

Let $\mathbf{f} = (f_{-1}, f_0, f_1, \dots, f_{d-1}) \in \mathbb{N}_0^{d+1}$ be a sequence of nonnegative integers. Then the following conditions are equivalent.

- (i) The sequence $\mathbf{f}$ is the f -vector of a simplicial complex of dimension at most $d-1$.
- (ii) The family $\mathcal{F}(\mathbf{f}) := \{F_j(k) : 0 \leq k \leq d, 1 \leq j \leq f_{k-1}\}$ is a simplicial complex (that is, with every set it contains all subsets).
- (iii) $f_{-1} = 1$, and $f_{k-1} \geq \partial_{k+1}(f_k)$ for $0 \leq k \leq d-1$.

Proof. The implication (ii) $\implies$ (i) is trivial, and the equivalence (ii) $\iff$ (iii) is clear with our construction of the “boundary operator” $\partial_k(n)$ above.

The remaining nontrivial part is (i) $\implies$ (ii): see Greene & Kleitman [241, p. 73] for a nice and simple proof by “compression.” $\square$

A simplicial complex (on a vertex set $V \subseteq \mathbb{N}$) is *compressed* if its k -faces form an initial segment with respect to r -lex order, for all k , that is, if it is a complex as given by Theorem 8.32(ii).

The compression technique mentioned for the last proof takes as an input a simplicial complex, and outputs a compressed simplicial complex with the same f -vector. The technique stems from a paper by Lindström & Zetterström [364]. It works quite the same way for multicomplexes (see Macaulay’s theorem 8.34 below), and also for a generalization of both theorems, due to Clements & Lindström [160] [18, Sect. 9.1].

What we really need for the following is not this theorem for simplicial complexes, but a version for “multicomplexes.” For this, we introduce some new terminology — I guess you’ve seen some of this before, but perhaps with different names.

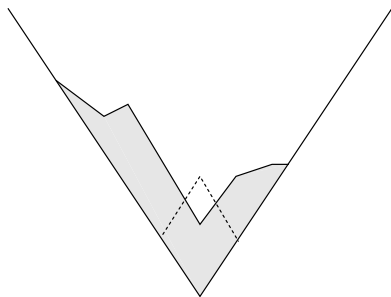
A *multiset* is a finite sequence of elements that may contain repeated elements. The order of the elements is irrelevant, but their multiplicities

are part of the structure. So, a multiset $\tilde{F}$ with elements in $\mathbb{N}$ can be uniquely written in the form

$$\tilde{F} = \{b_1, b_2, \dots, b_{k-1}, b_k\}_{\leq},$$

where the subscript “ $\leq$ ” indicates that we have arranged the elements in weakly increasing order, $b_1 \leq b_2 \leq \dots \leq b_k$. The *size* of a multiset is the number of elements, counting multiplicities. So the multiset $\tilde{F}$ above has size $|\tilde{F}| = k$, and we would call it a *k-multiset*. Also, a *submultiset* $\tilde{G} \subseteq \tilde{F}$ is a multiset in which every element has smaller or equal multiplicity than in $\tilde{F}$. Finally, a *multicomplex* is a finite collection of multisets that is closed under taking submultisets.

Multicomplexes can be interpreted in a variety of different ways (see Exercise 8.22). For example, they are equivalent to order ideals in $\mathbb{N}_0^n$ and to systems of monomials that are closed under taking factors. Our sketch shows the “generic” drawing of what a multicomplex might look like. Note the small diamond shape at the bottom, which denotes all the sets in the multiset system.



One can attempt a (quite technical) topological interpretation of multicomplexes, leading to the extensive apparatus of *semisimplicial sets* [386] — which we avoid. There are only a few pieces of topological terminology we use. So, the *dimension* of a multiset is defined to be one less than its size, $\dim(\tilde{F}) := |\tilde{F}| - 1$; the dimension of a multicomplex is the greatest dimension of a multiset it contains; and the *f-vector* of a multicomplex is $(f_{-1}, f_0, f_1, \dots, f_d)$, where f_i is the number of multisets of dimension i in the multicomplex.

Here is a basic bijection, which takes the k -multisets with elements from $[n]$ to the k -subsets of $[n+k-1]$:

$$\phi: \{b_1, \dots, b_k\}_{\leq} \longmapsto \{b_1, b_2 + 1, \dots, b_k + k - 1\}_{<}. \quad (8.33)$$

In particular, ϕ proves the basic identity

$$\binom{n}{k} = \binom{n+k-1}{k},$$

where the symbol on the left side denotes the number of k -multisets with elements from $[n]$ — the multiset analogue of the binomial coefficient $\binom{n}{k}$. See Exercise 8.22 for three other proofs of this.

Many set concepts are easily generalized to multiset concepts, if we just replace binomial coefficients by their multiset counterparts*. In particular, we need the *r-lex order* on k -multisets. For this we write

$$\tilde{F} \prec \tilde{G}$$

if $\max(\tilde{F}) < \max(\tilde{G})$, or if $\max(\tilde{F}) = \max(\tilde{G}) =: p$ and $\tilde{F} \setminus p \prec \tilde{G} \setminus p$, where “ $\tilde{F} \setminus p$ ” means that we remove exactly one copy of the largest element from $\tilde{F}$. So, r-lex is a linear order on the set of all k -multisubsets of $\mathbb{N}$. All the nice properties of r-lex order on sets generalize to multisets. The reason is that under the bijection 8.33, r-lex order on k -multisets is equivalent to r-lex order on k -sets,

$$\tilde{F} \prec \tilde{G} \iff \phi(\tilde{F}) \prec \phi(\tilde{G}).$$

Thus, for every k -multisubset, there is only a finite number of smaller ones, and thus we can use r-lex order to enumerate and label the k -multisubsets of $\mathbb{N}$, as $\tilde{F}_1(k), \tilde{F}_2(k), \dots$. Thus we define $\tilde{F}_j(k)$ to be the j th multiset in the listing according to r-lex order, and find that in fact it is the ϕ -image of the j th subset:

$$\phi(\tilde{F}_j(k)) = F_j(k).$$

For example, the r-lex order on the 3-multisubsets of $\mathbb{N}$ begins

$$\tilde{F}_1(3) \prec \tilde{F}_2(3) \prec \tilde{F}_3(3) \prec \tilde{F}_4(3) \prec \tilde{F}_5(3) \prec \tilde{F}_6(3) \prec \tilde{F}_7(3) \prec \tilde{F}_8(3) \prec \dots,$$

that is,

$$111 \prec 112 \prec 122 \prec 222 \prec 113 \prec 123 \prec 223 \prec 133 \prec \dots$$

Now, for $n, k \geq 0$ there is a unique expansion of n of the form

$$n = \binom{b_k}{k} + \binom{b_{k-1}}{k-1} + \dots + \binom{b_2}{2} + \binom{b_1}{1}$$

with $b_k \geq b_{k-1} \geq \dots \geq b_2 \geq b_1 \geq 0$

— we get this the same way as before, by defining

$$\tilde{F}_{n+1}(k) =: \{b_1 + 1, b_2 + 1, \dots, b_{k-1} + 1, b_k + 1\}_\leq,$$

or by setting $b_i := a_i - i + 1$ in the expansion we had before.

*Check the following carefully! Do not take my word for it! Go ahead! And don't ask Helga!

There are exactly n different k -multisets that are smaller in r-lex order than $\{b_1 + 1, \dots, b_k + 1\}$. Namely, for $\binom{b_k}{k}$ of them the maximal element is smaller than $b_k + 1$; for $\binom{b_{k-1}}{k-1}$ the maximal element is $b_k + 1$ but the next element is smaller than $b_{k-1} + 1$; and so on.

One more thing is easy to see:* the $(k-1)$ -multisets that are contained in some $\tilde{F}_j(k)$ with $j \leq n+1$ also have maximal element smaller than $b_k + 1$, or they have maximal element $b_k + 1$ but the next smallest element smaller than $b_{k-1} + 1$, and so forth — so there are exactly

$$\begin{aligned} \partial^k(n+1) &:= \binom{b_k}{k-1} + \binom{b_{k-1}}{k-2} + \dots + \binom{b_2}{1} + \binom{b_1}{0} \\ &= \binom{b_k + k - 2}{k-1} + \binom{b_{k-1} + k - 3}{k-2} + \dots + \binom{b_2}{1} + \binom{b_1 - 1}{0} \\ &= \binom{a_k - 1}{k-1} + \binom{a_{k-1} - 1}{k-2} + \dots + \binom{a_2 - 1}{1} + \binom{a_1 - 1}{0} \end{aligned}$$

of them.

For an example, again let $k = 3$ and $n = 7$. We can expand

$$7 = \binom{2}{3} + \binom{2}{2} + \binom{0}{1},$$

and from this we see that $\tilde{F}_8(3) = \{1, 3, 3\}$, as in the listing above. There are 7 smaller 3-multisets in r-lex order, where $\binom{2}{3} = 4$ have largest element smaller than 3, $\binom{2}{2} = 3$ have largest element 3 but the next element smaller than 3, and $\binom{0}{1} = 0$ have the two largest elements equal to 3 but the smallest element smaller than 1. Also, there are

$$\partial^3(8) = \binom{2}{2} + \binom{2}{1} + \binom{0}{0} = \binom{3}{2} + \binom{2}{1} + \binom{-1}{0} = 3 + 2 + 1 = 6$$

2-multisets contained in the first eight 3-sets, namely $3 = \binom{2}{2}$ with largest element smaller than 3, $2 = \binom{2}{1}$ with largest element 3 but the next one smaller than 3, and $1 = \binom{-1}{0}$ with the elements 3 and 3. This last one is contained in $F_8(3)$, but not in a smaller 3-set.

Our main reason of doing multisets and their r-lex ordering is to get some intuition for what multicomplexes are, how they behave — to be able to make sense out of the following theorem. It uses a “relative” Φ_d of the ϕ -map 8.33, which takes a multiset $\{b_k, \dots, b_1\}_{\geq}$, adds 1 to each of the elements, adjoins $d-k$ zeroes to the multiset, and then applies the ϕ -map to get a set:

$$\begin{aligned} \Phi_d(\{b_k, \dots, b_1\}_{\geq}) &:= \\ &= \phi(\{b_k + 1, \dots, b_1 + 1, \overbrace{0, 0, \dots, 0}^{d-k}\}_{\geq}) \\ &= \{b_k + 1 + d, b_{k-1} + 1 + d - 1, \dots, b_1 + 1 + d - k + 1, d - k, \dots, 2, 1\}_{>} \end{aligned}$$

*Do you get a déjà-vu feeling? Of course, what we are doing here for multisets is exactly the same as we did for sets before!

Theorem 8.34 (Macaulay's theorem).

Let $\mathbf{h} = (h_0, h_1, \dots, h_d) \in \mathbb{N}_0^{d+1}$ be a sequence of nonnegative integers. Then the following are equivalent.

- (i) The sequence $\mathbf{h}$ is the f -vector of a multicomplex.
- (ii) The sequence $\mathbf{h}$ is the f -vector of a compressed multicomplex, that is, $\tilde{\mathcal{F}} := \{\tilde{F}_j(k) : 0 \leq k \leq d, 1 \leq j \leq h_k\}$ is a multicomplex.
- (iii) $h_0 = 1$, and $h_{k-1} \geq \partial^k(h_k)$ for $1 \leq k \leq d$.
- (iv) The sequence $\mathbf{h}$ is the h -vector of a shellable simplicial complex of dimension $d-1$.
- (v) The family $\{\Phi_d(\tilde{F}_j(k)) : 0 \leq k \leq d, 1 \leq j \leq h_k\}$ is the set of facets of a shellable simplicial complex with h -vector $\mathbf{h}$.

Proof. Again part (ii) $\implies$ (i) is trivial, while (ii) $\iff$ (iii) follows from our previous discussion.

The part (i) $\implies$ (ii), *from multicomplexes to compressed multicomplexes*, is originally due to Macaulay. It can be proved by the “compression” technique that we have mentioned in the proof of the Kruskal-Katona Theorem 8.32.

For (ii) $\implies$ (v), *from multicomplexes to shellable complexes*, this is the special case “ $s = 1$ ” of a construction in Björner, Frankl & Stanley [93], which takes a multicomplex and produces a pure complex from it:

$$\hat{\Phi}_d : \tilde{\mathcal{F}} \longrightarrow \{\tilde{G} : \tilde{G} \subseteq \Phi_d(\tilde{F}) \text{ for some } \tilde{F} \in \tilde{\mathcal{F}}\}.$$

If $\tilde{\mathcal{F}}$ is the compressed multicomplex from (ii), then the pure complex $\hat{\Phi}_d(\tilde{\mathcal{F}})$ is shellable. In fact, in this case r -lex order defines a shelling, and the restriction set is $R(\Phi_d(\tilde{F}_j(k))) = \Phi_d(\tilde{F}_j(k)) \setminus \{d-k, \dots, 2, 1\}$, of cardinality k . Thus every k -multiset in the multicomplex contributes “1” to h_k in the h -vector of the corresponding simplicial complex. For the proof with details we refer to [93, Sect. 5].

The implication (v) $\implies$ (iv) is trivial, thus we are left with proving the direction (iv) $\implies$ (i), *from shellable complexes to multicomplexes*. For this, Stanley [512, 513] has given an algebraic argument: the multicomplex arises in this case from a monomial basis for “the Stanley-Reisner ring modulo a system of parameters.” Is there a simple combinatorial argument? Note that this innocent-looking claim in particular implies that

$$h_k \leq \left(\binom{h_1}{k} \right) = \binom{h_1 + k - 1}{k}$$

for shellable complexes, and thus this reproves the upper bound theorem (McMullen's Lemma 8.26)! In fact, this is the key to Stanley's proof of the upper-bound conjecture for spheres [511] [515, Sect. II.3]. $\square$